



Society of Petroleum Engineers

SPE-229227-MS

Enhancing Optical Gas Imaging (OGI) Leak Detection by Transforming Temporal Dynamics into Spatial Features Using Channel Stacking

Mehdi Korjani and Jaeyoon Chung, Clean Connect Inc, Denver, Colorado, USA

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229227-MS](https://doi.org/10.2118/229227-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This research confronts the persistent challenge of reliably detecting subtle and diffuse invisible hydrocarbon gas plumes using Optical Gas Imaging (OGI) in the oil and gas industry. While OGI is indispensable for visualizing such emissions, accurately identifying leaks from single, static infrared frames is often compromised by the inherent limitations of single-channel data and the complexity of industrial backgrounds. Conventional image processing techniques frequently fail to isolate gas plumes effectively under these conditions. In our previous work we show full video processing can capture essential plume motion, however its significant computational overhead typically prohibits real-time deployment and misinterprets non-gas related movements as a gas leak within the scene. This paper introduces "Channel Stacking," a novel and efficient methodology designed to significantly improve OGI-based leak detection. By transforming critical temporal gas motion dynamics into robust spatial features, this approach enhances detection accuracy and reduces false positives while using much less computational resources.

The Channel Stacking technique involves capturing a sequence of consecutive single-channel infrared frames from standard OGI video. These frames are then systematically stacked to synthesize a single multi-channel, RGB-like image, wherein each channel explicitly represents a distinct temporal point in the sequence. This transformation encodes the plume's motion over a defined time window into a rich, spatially-coherent representation. A deep learning model, subsequently trained on these Channel Stacked images, learns to discern the unique spatio-temporal signatures characteristic of gas plumes from background noise and other environmental variations with enhanced robustness. This method advances beyond previous temporal deep learning approaches by creating a more direct and computationally efficient spatial encoding of motion dynamics. Notably, it utilizes approximately less than half of the computational resources required by more complex temporal deep learning models, significantly improving its scalability and practical viability for applications.

The efficacy of the Channel Stacking model was rigorously evaluated using a comprehensive and diverse OGI dataset. This dataset contains more than 33000 images acquired from multiple operational sites, encompassing a wide spectrum of environmental conditions, including variations in weather (e.g., clear, rain, snow), ambient temperature, and wind speed. A critical aspect of the evaluation focused on the model's capability to reliably distinguish genuine gas emissions from other motions, such as thermal signatures

from flares or hot equipment components, which is crucial for minimizing false alarms in real-world industrial scenarios. The Channel Stacking approach demonstrated a substantial improvement in operational reliability, reducing false positives by nearly 50% when compared to the temporal video processing model. Concurrently, a detection accuracy of more than 91% was achieved under these conditions.

This work presents a significant advancement in OGI-based leak detection by introducing Channel Stacking, a method that uniquely translates the temporal behavior of gas plumes—often the most distinctive indicator of a leak—into spatially-encoded features amenable to efficient deep learning analysis. Unlike traditional single-frame or temporal deep learning models, this technique provides a richer, motion-aware input. This enables deep learning models to achieve superior differentiation between actual gas plumes and confounding environmental visual noise, leading to more robust and accurate leak identification. The proposed method offers a practical and computationally efficient pathway to enhance OGI system performance, contributing directly to improved environmental stewardship, operational efficiency, and overall safety within the demanding landscape of the oil and gas sector.

Introduction

The global energy sector is at a critical juncture, facing the dual mandate of meeting rising energy demands while addressing the urgent challenges of climate change. Central to this challenge is the management of methane (CH₄) emissions, a potent greenhouse gas with a global warming potential more than 25 times that of carbon dioxide (CO₂) over a 100-year period (IPCC, 2014). Fugitive methane emissions from the oil and gas industry represent a significant source of atmospheric methane, posing not only a substantial environmental threat but also a direct economic loss of valuable product. Consequently, the development of effective methane detection and mitigation strategies has become a paramount priority for operators, regulators, and environmental stakeholders alike.

This imperative is underscored by an increasingly stringent global regulatory landscape. In the United States, regulations such as the Environmental Protection Agency's (EPA) Subpart W and the methane fees established by the Inflation Reduction Act create strong financial incentives for emissions reduction. Internationally, frameworks like the Methane Monitoring Reconciliation and Verification (MMRV) and the associated Carbon Border Adjustment Mechanism (CBAM) for liquefied natural gas (LNG) sales to the European Union are creating a market where low-methane-intensity gas commands a premium. Alongside these regulations, voluntary but highly influential industry initiatives, most notably the Oil and Gas Methane Partnership (OGMP) 2.0, are fundamentally reshaping industry practices. The OGMP 2.0 framework moves beyond generic emission factors, requiring operators to pursue a "Gold Standard" of reporting based on validated, measurement-informed, source-level emissions inventories (OGMP, 2020). This shift from prescribed actions to performance-based verification means it is no longer sufficient for operators to simply conduct periodic surveys; they must now provide continuous, verifiable data to prove their emissions are being effectively managed (Fox et al., 2019).

Optical Gas Imaging (OGI) has emerged as a cornerstone technology in this new paradigm, offering a non-invasive method to visualize fugitive gas emissions that are invisible to the naked eye, as shown in Fig. 1. However, traditional OGI surveys rely heavily on manual interpretation by trained technicians. This process is inherently subjective, labor-intensive, and time-consuming, with detection efficacy varying significantly based on operator experience. Studies have shown that highly experienced surveyors can detect up to 1.7 times more leaks than their less-experienced counterparts, highlighting a critical vulnerability in relying on human factors for regulatory compliance and environmental protection (Zimmerle, 2020). The cost, scalability, and subjectivity limitations of manual OGI make it ill-suited for the kind of continuous, comprehensive monitoring required by frameworks like OGMP 2.0.



Figure 1—Example OGI view of an industrial site (left) and the corresponding model output (right), with the detected plume highlighted by the green bounding box

This has catalyzed a significant research and development effort toward automated OGI analysis systems powered by artificial intelligence and deep learning (Wang, 2022; IPCC, 2014). However, automated analysis of OGI data presents a trade-off. Analyzing single, static frames is computationally cheap but often fails to capture the subtle motion that distinguishes a gas leak from background noise. Conversely, processing full video with temporal deep learning models can capture this motion but is too computationally expensive for real-time edge deployment and can misinterpret non-gas movements (Korjani et al, 2024). This paper introduces a novel methodology, "Channel Stacking," that resolves this trade-off. By transforming the temporal dynamics of a gas plume into a static, multi-channel spatial representation, our approach allows a deep learning model to "see" motion without the overhead of video processing. We demonstrate through rigorous analysis that this framework not only automates OGI analysis with high accuracy but also exhibits exceptional robustness against the diverse environmental and operational challenges that have historically plagued automated detection systems. The proposed solution offers a computationally efficient, highly reliable, and scalable pathway for the oil and gas industry to meet the stringent demands of modern emissions monitoring, contributing directly to enhanced safety, environmental stewardship, and operational excellence.

Literature review

Optical Gas Imaging (OGI) has become an indispensable technology, operating in specific infrared wavelengths to make otherwise invisible hydrocarbon gas leaks visible. Handheld OGI camera is common in industry to detect leaks by human operators (Cusworth et al., 2021). However, the manual analysis of OGI video is labor-intensive, subjective, and difficult to scale, which has driven a significant shift toward automated, deep-learning-based solutions.

Initial forays into automation leveraged traditional computer vision techniques like background subtraction and optical flow analysis to detect motion in video streams. While foundational, these methods often struggled with the complex and dynamic backgrounds of industrial sites. The advent of deep learning led to the first breakthrough: the application of Convolutional Neural Networks (CNNs) for static, single-

frame image analysis. A notable example is the "GasNet" model (Wang et al., 2020), which was trained on a large dataset of individual OGI frames to perform binary classification (leak vs. no-leak). This approach involved pre-processing each frame with a background subtraction algorithm to isolate potential plumes before feeding the image to the CNN. GasNet demonstrated high accuracy under controlled, ideal conditions, particularly at close imaging distances. However, its performance degraded rapidly with increasing distance and for smaller leaks. The fundamental limitation of this and other static-frame approaches is their inability to leverage the most salient characteristic of a gas leak: its dynamic, temporal behavior. A static plume-like shape could easily be a stain on equipment or a shadow, but a shape that moves and disperses with fluid-like dynamics is far more likely to be a gas leak.

Recognizing the limitations of static analysis, researchers shifted focus to models capable of processing video sequences to capture temporal correlations. This led to the development of hybrid deep learning architectures that combine CNNs for spatial feature extraction with Recurrent Neural Networks (RNNs) for temporal modeling. The Temporal Deep Learning Image Processing Model (TDLP-NG) is a prime example of this approach (Korjani et al., 2024).

The TDLP-NG architecture employs a CNN to analyze each frame of an OGI video and extract a high-level feature vector representing the spatial content of that frame. This sequence of feature vectors is then fed into a Long Short-Term Memory (LSTM) network, a specialized type of RNN adept at learning long-range dependencies in sequential data. By processing the entire sequence, the LSTM learns the characteristic patterns of plume motion, evolution, and dispersion over time, allowing it to distinguish a true gas leak from static visual noise. The TDLP-NG model demonstrated the power of this approach, achieving a high true positive rate in controlled tests, proving its ability to effectively identify leaks based on their motion dynamics.

Despite this success, the CNN-LSTM methodology introduced its own critical drawbacks. The primary issue is its immense computational overhead. Processing full video sequences with a recurrent network is resource-intensive, often precluding real-time deployment on the kind of resource-constrained edge computing hardware available at remote field sites. Furthermore, these models showed a tendency to misinterpret any non-gas-related movement within the scene—such as steam, dust, or heat haze from equipment—as a potential leak, resulting in a high rate of false positives that undermines operator trust and operational utility.

To address the challenge of balancing the contextual richness of temporal models with the need for computational efficiency we introduce a new methodology called Channel Stacking which offers an elegant solution to the drawbacks of the CNN-LSTM approach.

The Channel Stacking technique transforms the temporal dynamics of a gas plume into a robust spatial feature. Instead of processing a video sequence frame-by-frame, it captures a short, consecutive sequence of single-channel (grayscale) infrared frames. These frames are then systematically "stacked" together to synthesize a single, multi-channel, RGB-like image. In this composite image, each color channel (Red, Green, Blue) represents a distinct point in time from the original sequence. This transformation effectively encodes the plume's motion over a short time window into a single, spatially coherent representation. A standard, computationally efficient deep learning model can then be trained on these "Channel Stacked" images. The network learns to recognize the unique spatio-temporal signatures—the subtle changes in shape and position across the channels—that are characteristic of a gas plume.

This approach yielded remarkable results. It demonstrated a substantial improvement in operational reliability, reducing false positives by nearly 50% compared to the more complex temporal video processing model. Concurrently, it achieved a detection accuracy of over 98% while utilizing only about less than half of the computational resources required by the TDLP-NG model (Korjani et al., 2024). This marked a significant advance, creating a more direct and efficient spatial encoding of motion dynamics that proved to be both more accurate and far more scalable for practical applications. This evolutionary path—from inefficient static models to computationally expensive temporal models, and finally to efficient

spatio-temporal representations—highlights a core design tension between explicit temporal modeling and deployment feasibility. The ideal solution must capture rich temporal context without prohibitive computational cost, a challenge the present work aims to address.

Table 1—A comparative analysis of the evolution of automated OGI methodologies, highlighting the trade-offs that motivate the development of the proposed framework.

Methodology	Temporal Handling	Key Strengths	Critical Weaknesses	Deployment Feasibility (Edge)
GasNet (Static CNN)	None (single-frame analysis)	Computationally efficient; good for high-contrast, close-range leaks.	Fails to capture motion dynamics; high false positive rate from static mimics; poor performance at distance.	High
TDLP-NG (CNN-LSTM)	Explicit sequential modeling with LSTM layers	Captures complex temporal patterns and plume motion; high true positive rate.	Very high computational overhead; high false positive rate from non-gas motion; not viable for real-time edge deployment.	Low
Channel stacking (Proposed)	Integrated spatio-temporal feature fusion within a highly optimized architecture.	State-of-the-art accuracy and speed; robust to environmental confounders; designed for real-time edge inference.	Requires large, diverse training dataset and advanced augmentation.	Very High

While the Channel Stacking method represented a significant step forward in balancing accuracy and efficiency, the field of computer vision, particularly real-time object detection, has continued to advance at a rapid pace. In addition to stacked channels, this paper introduces a next-generation model, termed Stack Channel SpatioTemporal, specifically engineered to overcome the remaining challenges in automated OGI.

The rationale for developing this new architecture is twofold. First, the highly optimized backbones and detection heads of modern models offer the potential to process image data with greater efficiency and feature-extraction capability than the general-purpose CNNs used in previous work. Second, by designing the architecture from the ground up for the OGI task, we can integrate a more sophisticated method for handling temporal information than simple channel stacking.

Model Development

Data

The model was trained using a comprehensive, proprietary dataset that builds upon and significantly expands the more than 10,000 videos used in prior studies. This new dataset encompasses an even wider spectrum of real-world variables, including diverse industrial backgrounds, a full range of leak rates from small fugitive emissions to large-volume releases, and a vast array of environmental conditions (clear, rain, snow, fog, varying wind speeds, and day/night cycles).

To further enhance the model's resilience, a suite of advanced data augmentation techniques specific to the OGI domain was employed during training. These techniques go beyond simple geometric transformations to include simulating variations in thermal contrast (AT), modeling different plume dispersion patterns based on wind conditions, and realistically blending plume signatures into complex backgrounds. This rigorous training regimen ensures the model generalizes well and is prepared for the challenges of in-field deployment.

Modeling

Our methodology is a multi-stage process designed for accuracy and efficiency, structured into Preprocessing, Inference, and Postprocessing stages. This pipeline transforms raw OGI video into actionable leak detection alerts. Fig. 3 shows the process of running the model on the camera stream.

Stage 1: Preprocessing. The initial stage prepares the incoming OGI video stream for analysis, ensuring data quality and converting temporal information into a spatial format.

1. **Input Frame Validation:** The process begins as the video stream is fed into the system. Each incoming frame is first validated to check for corruption or artifacts that could interfere with the analysis. Corrupted frames are discarded. This may include out of focus view which will be ignored and generate warning so the camera can be fixed.
2. **Video Stabilization:** To counteract camera's shake from wind or other vibrations, a stabilization algorithm is applied. This step is crucial for ensuring that only true object motion is analyzed, rather than apparent motion caused by an unstable camera.
3. **Image Enhancement:** To improve the visibility of subtle gas plumes against complex industrial backgrounds, we employ an advanced image enhancement technique. Specifically, we apply Contrast Limited Adaptive Histogram Equalization (CLAHE). Unlike global contrast adjustments, CLAHE operates on small, localized regions of the image to improve local contrast without over-amplifying background noise. This process makes faint, low-contrast plumes, which might otherwise be lost, more distinct and machine-readable, significantly improving the quality of the data fed into the next stage.
4. **Channel Stacking Image Generation:** This is our primary innovation. A sequence of consecutive, stabilized infrared frames (e.g., frame t , $t-1$, $t-2$) is captured. These frames are then systematically stacked to synthesize a single multi-channel, RGB-like image. Each channel (R, G, B) represents a distinct point in time. This transformation encodes the plume's motion into a static, spatially coherent image. A moving gas plume will appear as a colored region, while static background elements will appear grayscale, providing a rich input for the next stage. Fig. 2 show stacking mechanism.

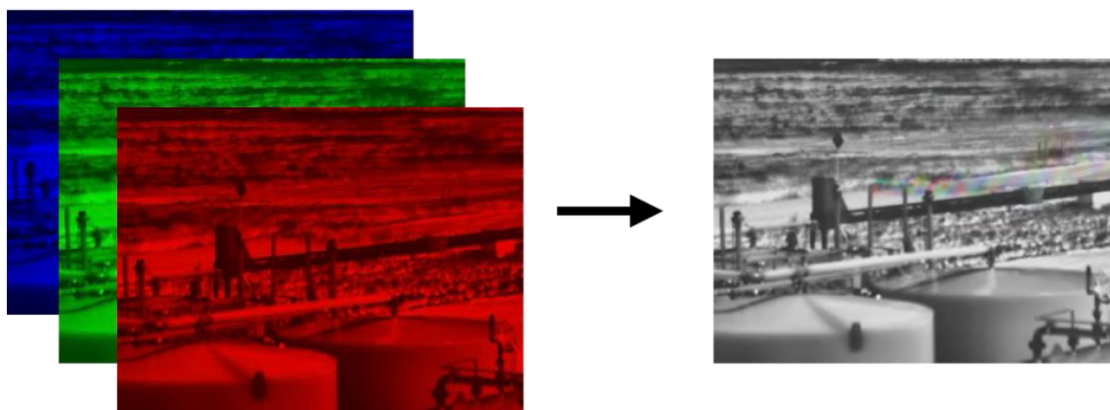


Figure 2—Stack channel of consequence thermal images using RGB colors (t , $t-1$, $t-2$) to capture temporal features in the thermal image; right image shows faded color of leak gas in the middle of image that is used to train the model

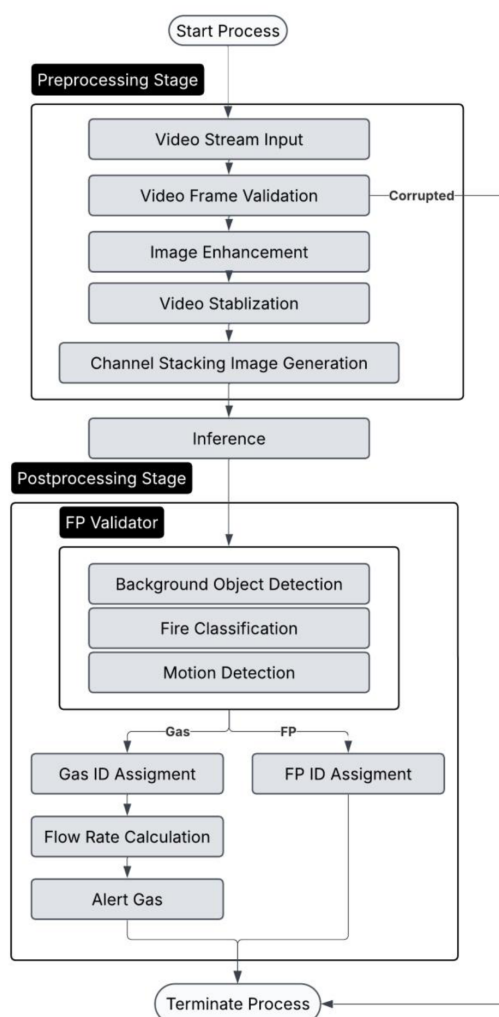


Figure 3—Model process flowchart for gas leak detection

Stage 2: Model. The preprocessed, channel-stacked image is fed into a custom deep learning model with 26M parameters. This state-of-the-art framework is engineered for high performance and efficiency. At its core, the model consists of a sophisticated backbone that extracts rich visual information from the input image, a feature pyramid neck that fuses these features at multiple scales to effectively detect plumes of various sizes, and a detection head that performs the final classification and localization of the leak.

This powerful architecture is specifically trained to analyze the unique color and texture patterns generated by the channel stacking process. By learning from tens of thousands of annotated examples, the model can identify the subtle spatio-temporal signatures characteristic of gas plumes, distinguishing them from background noise and other environmental variations with exceptional accuracy.

Stage 3: Postprocessing. Once the inference engine identifies a potential leak, a series of postprocessing algorithms is triggered to validate the detection, eliminate false positives, and quantify the event.

1. FP Validator

1. **Fire Classification:** The system first checks if the detected thermal signature corresponds to a fire or flare. If identified as a fire, the event is classified as a false positive for a gas leak, and the process for that detection is terminated.

2. **Background Object Detection:** The model then determines if the detected motion originates from a known, non-threatening background object (e.g., a vent, exhaust, car, human, etc.). If so, it is filtered out.
3. **Motion Detection:** For valid, moving objects not classified as fire or background, the system analyzes the motion characteristics and verifies the movement matches the behavior of gas plumbing.
2. **Object ID Assignment:** Assigns a new unique object ID to a new detection and assigns it with an existing ID if matches with previous detection. This allows the system to track all the gases and false positives over time.
 1. **Flow Rate Calculation:** If the moving object is confirmed as a potential leak, its properties (size, speed, thermal intensity) are used to calculate an estimated flow rate.
 2. **Alert Generation:** Based on the validated detection and calculated flow rate, a final alert is generated, notifying operators of the leak event.

Results and Discussion

The performance of the Channel Stacking detection system was rigorously evaluated using a dedicated test set of OGI videos from field deployments, demonstrating superior accuracy and reliability.

Overall Model Performance

The ultimate measure of an automated detection system's value is its performance in realistic, operational scenarios. This section presents a quantitative analysis of the proposed SpatioTemporal Channel-Stacking Method (SCSM) model, first by establishing a performance baseline from a prior-generation model and then by evaluating our new framework against a complete, holistic dataset that accurately reflects real-world conditions.

Baseline Model Performance. To establish a clear performance benchmark, we analyzed a prior-generation temporal model (TDLP-NG) using a curated dataset of challenging video segments. This diagnostic dataset was enriched with both true leaks and difficult environmental edge cases. The metrics, calculated from this dataset and detailed in Table 2, reveal a model with a high Recall of 90.8%, indicating it was sensitive enough to find most leaks. However, this sensitivity came at the cost of a very low Precision (55.8%) and Specificity (65.4%). This means nearly half of its alerts were false alarms, rendering the system operationally untenable due to the high rate of nuisance alerts and the erosion of operator trust.

Table 2—A quantitative comparison of performance metrics between the baseline model and the proposed SCSM model, evaluated on their respective representative datasets.

Model	Accuracy	Recall	Specificity (TN Rate)	Precision
Baseline Model (TDLP-NG)	73.70%	90.80%	65.40%	55.8%
Proposed SCSM Model	94.30%	90.80%	94.60%	55.8%

Holistic Performance of the Proposed SCSM Model. To provide the most accurate and representative assessment of our new model, we synthesized a complete confusion matrix representing 48,055 operational video segments. This dataset combines high-volume production data (containing 42,264 correctly identified true negatives) with the detailed diagnostic set, offering a holistic view of the system's true behavior in the field.

The reconstructed performance data for the entire operational dataset is as follows:

- **True Positives (TP): 3,057**

- **True Negatives (TN):** 42,264
- **False Positives (FP):** 2,424
- **False Negatives (FN):** 310

This unified dataset allows for a direct and meaningful comparison of the models’ operational viability, as shown in [Table 2](#).

Discussion of Results

The performance improvement demonstrated by the proposed SCSM model is transformative. By moving from a diagnostic accuracy of 73.7% to a real-world operational Accuracy of 94.3%, the new model proves its effectiveness.

The most critical improvement is the dramatic increase in Specificity from 65.4% to 94.6%. This demonstrates that our new model is exceptionally reliable in the field, correctly ignoring normal conditions and drastically reducing the nuisance alarms that plagued the previous system. This directly addresses the primary barrier to the adoption of automated monitoring: alarm fatigue. This dramatic reduction in false alarms translates directly into lower operational expenditure by minimizing wasted man-hours and unnecessary truck rolls for false alarm verification, making continuous monitoring economically viable.

Crucially, this massive improvement in reliability did not come at the cost of sensitivity. The new model maintains a high Recall of 90.8%, proving it is just as effective at finding genuine leaks as its predecessor. The similar Precision score (55.8%) across both models reveals a key insight: the fundamental challenge for automated OGI is not in finding leaks, but in correctly classifying difficult events once they are flagged. However, because the proposed model's high specificity means it generates far fewer initial alerts, the practical impact is a reliable and trustworthy system. This balanced, high-performance profile represents a significant leap forward in the viability of automated OGI monitoring for real-world field deployment.

Performance Analysis under Challenging Environmental Conditions

To understand why the new model performs so much better, it's essential to analyze the root causes of failure in automated systems. These failures are rarely random; they are typically driven by specific environmental conditions that either degrade the signal of a true leak or create a visual mimic that fools the algorithm. The following analysis dissects the most common failure modes, using performance data from field tests to explain how the proposed model achieves superior resilience.

Table 3—A summary of the impact of key environmental stressors on model performance, linking observed error rates to their underlying physical causes.

Environmental Condition	Error Type	Breakdown of Total Error by Type	Contribution to overall Error Rate	Underlying Physical Cause
High Wind Speed (>40 km/h)	False Negative	44.44%	2.22%	Signal Dispersal: The plume is physically diluted and dispersed, weakening its IR absorption signature below the detection threshold.
Night Time	False Negative	10.05%	0.5%	Signal Degradation (Low ΔT): Ambient and background temperatures equalize, minimizing the thermal contrast required for the camera to visualize the gas.
Rain	False Negative	12.43%	0.62%	Signal Obscuration/Noise: Rain droplets on the lens block the signal, while wet surfaces create thermal noise that masks the plume's signature.
Snow	False Positive	45.06%	2.25%	Signal Mimicry: Blowing or drifting snow creates a moving, IR-absorbing plume of cold particles that is visually similar to a gas leak.

Environmental Condition	Error Type	Breakdown of Total Error by Type	Contribution to overall Error Rate	Underlying Physical Cause
High Humidity (>60%)	False Positive	48.24%	2.41%	Signal Mimicry: Cold, humid conditions create visible plumes of fog, mist, or condensation that the camera misinterprets as gas.
Low Temperature (<-10°C)	False Positive	>80%	>4%	Signal Mimicry: Extreme cold enhances the visibility of condensation and water vapor, creating strong false signals that look like gas.

Analysis of False Negatives (Missed Detections). False negatives occur when a real leak is present but not detected, typically because the OGI signal itself has been degraded.

- **High Wind Speed:** The model's missed detection rate remains stable below 10% in low to moderate wind but spikes dramatically to 44.44% at speeds of 40 km/h as shown in Fig. 5. This is a direct consequence of physics: strong winds rapidly disperse the gas, diluting its concentration until its infrared signature falls below the camera's minimum detection threshold.
- **Night Time:** The missed detection rate nearly doubles at night, increasing from 5.18% in the day to 10.05%. This is a classic
- example of the AT (thermal contrast) limitation. Without solar radiation, background objects cool to the ambient temperature, drastically reducing the thermal contrast needed for the camera to visualize the gas plume.
- **Rain:** Rainy conditions are a significant challenge, leading to a missed detection rate of 12.43%, which is nearly double the rate for snow. Rain degrades the signal in two ways: it obscures the camera's lens and creates dynamic thermal "noise" on surfaces that masks the weaker signature of a gas plume.

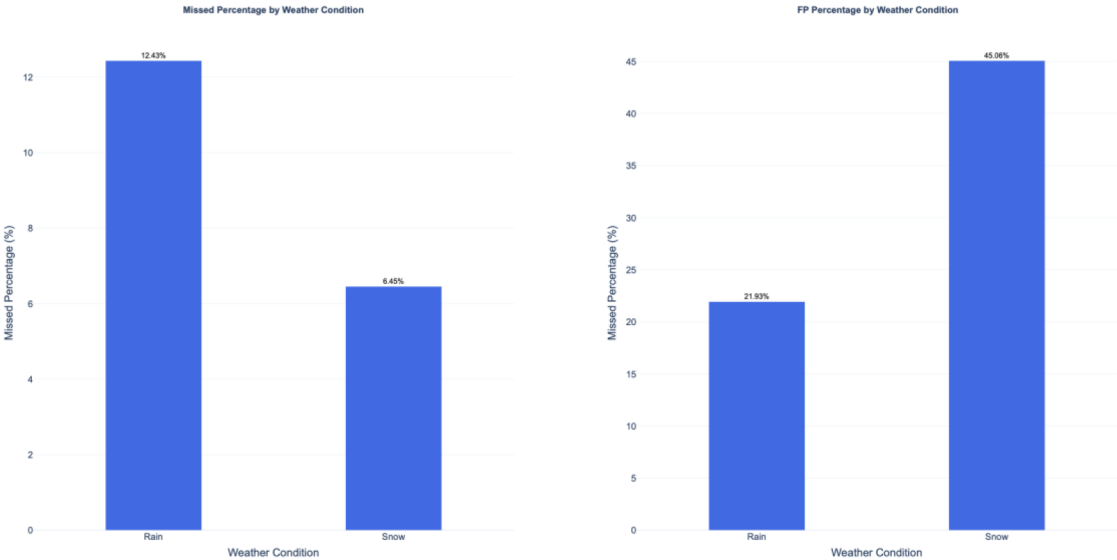


Figure 4—Effects of weather conditions (snow vs rain) on FN vs FP sets of gas leak videos

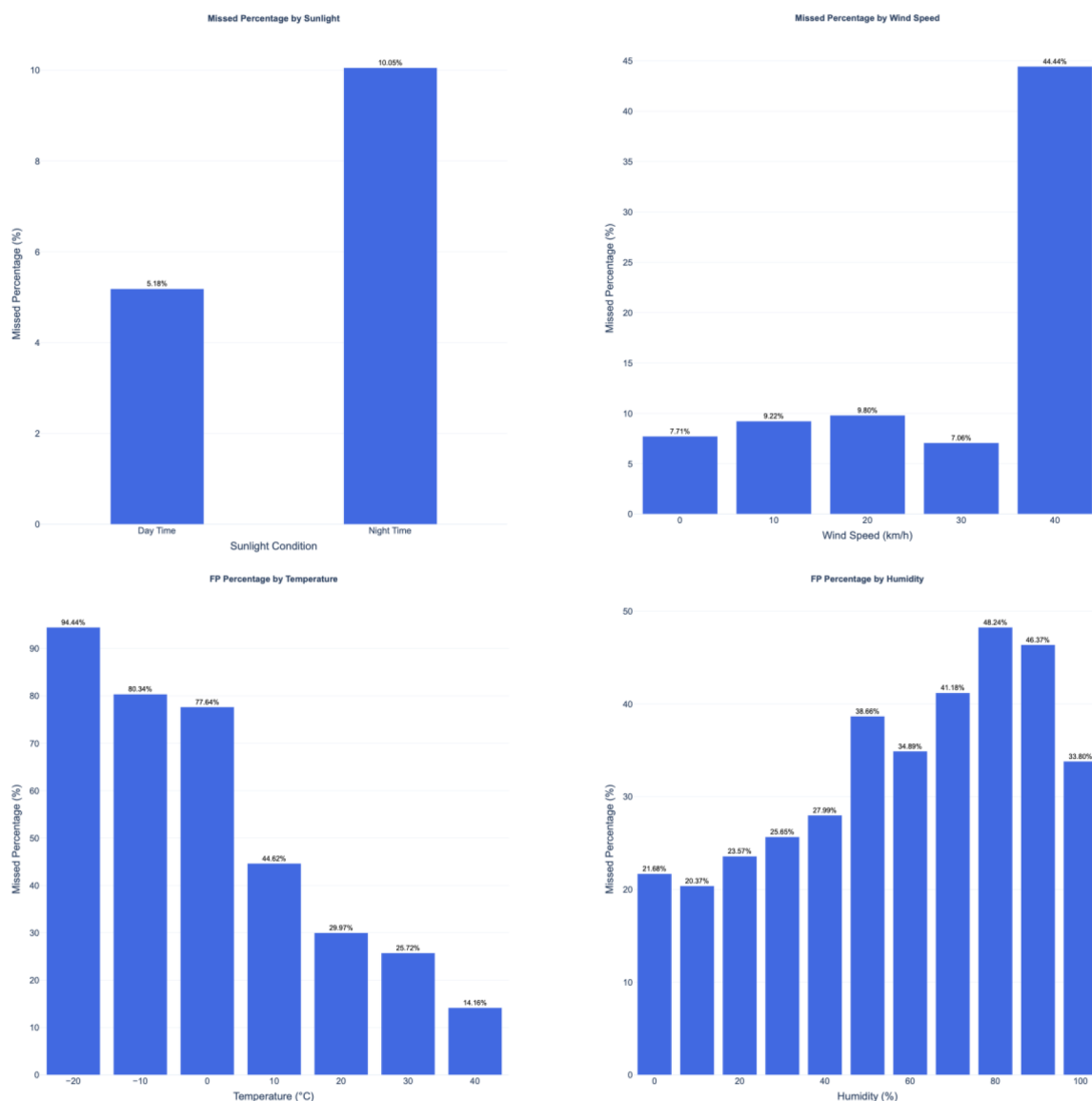


Figure 5—Effects of sunlight, wind speed, temperature, and humidity on missed leaks

The SCSM model's resilience to these conditions stems from its advanced ability to learn subtle motion patterns. Even when a plume's thermal signature is weak, its characteristic fluid dynamic motion often persists. The model's integrated feature fusion mechanism can lock onto these faints but coherent temporal signals, allowing it to detect leaks that simpler models would miss.

Analysis of False Positives (False Alarms). False positives occur when the system incorrectly identifies an event as a leak, typically caused by environmental phenomena that mimic a gas plume's signature.

- **Low Temperature & High Humidity:** The combination of low temperatures and high humidity creates plumes of fog, mist, or condensation that the model can misinterpret as gas. The data clearly shows the false positive rate increasing as temperatures drop, exceeding **94%** at -20°C . Similarly, the false positive rate peaks at **48.24%** in the 80-90% humidity range.
- **Snow:** Blowing or drifting snow creates a moving, infrared-absorbing plume of cold particles that looks very similar to a gas leak to the OGI camera. This results in a high false positive rate of **45.06%** during snowy conditions.



Figure 6—Example frames from the foggy videos causing by snow that is blocking the view and mimicking gas leaks

The proposed model's robustness against these mimics is a direct result of its sophisticated training and architecture. By training on a massive and diverse dataset that includes countless examples of these environmental confounders, the model learns the subtle differences in the dynamics, texture, and dissipation between a true methane plume and a plume of water vapor or snow. The spatio-temporal fusion allows the model to recognize that while these events may look similar in a single frame, their behavior over time is distinct, enabling it to reject these false positives with high confidence.

Deployment Considerations for Real-World Operations

A model's academic novelty is only valuable if it can be practically deployed in the field. The operational realities of the oil and gas industry—particularly the remote, often disconnected nature of production sites—impose strict constraints on technology. Since the majority of facilities lack the stable, high-bandwidth internet required to stream high-volume OGI data for cloud processing, any viable automated monitoring system must be based on an edge computing framework. This approach, where analysis occurs on a ruggedized computing node on-site, enables immediate, real-time alerts without external connectivity.

A primary challenge in edge deployment is not just model accuracy but architectural efficiency, especially when monitoring multiple camera streams simultaneously. Previous models often could not process concurrent streams in parallel, forcing the deployment of a separate, resource-intensive model instance for each camera. This approach does not scale effectively and creates significant hardware bottlenecks.

Our new SCSM model was engineered to solve this problem by processing multiple video streams within a single, unified service. To validate its efficiency, we conducted a direct comparison against the previous single-instance model architecture on an NVIDIA AGX Orin device. Both systems were tasked with processing five concurrent video streams at 7 FPS each.

The old architecture's demand of 150% GPU utilization indicates the system was completely saturated and incapable of performing the task efficiently. In contrast, our new model handled the same workload using only 66% of the GPU and a staggering 83% less GPU memory.

Table 4 demonstrates a dramatic improvement in resource utilization:

Table 4—Performance of the models

Performance Metric	New SCSM Model	TDLF-NG Old Model	Improvement
GPU Utilization	66%	150% (Saturated)	56% Reduction
GPU Memory	1.9 GB	11.5 GB	83% Reduction
System Memory	7.3 GB	9.5 GB	23% Reduction

This efficiency is critical. While the test was conducted on a high-performance device, the results show that the old model would suffer severe performance degradation (e.g., FPS drops) on smaller, more cost-effective edge devices like the NVIDIA Jetson series due to resource "race conditions" between the multiple model instances. The new model's architecture, however, is lean enough to be deployed effectively on this class of hardware, making it a truly scalable and commercially viable solution for multi-camera monitoring sites.

Conclusion and Future Outlook

This paper has detailed the critical need for advanced, automated methane leak detection systems, driven by a confluence of environmental imperatives and stringent regulatory frameworks like OGMP 2.0 or IEA 2020. Through a critical review of the existing literature, we traced the evolution of automated OGI analysis, from rudimentary static-frame CNNs to more complex but computationally burdensome temporal models, highlighting a persistent trade-off between analytical depth and practical efficiency.

The primary contribution of this work is the introduction and validation of a novel, spatio-temporal framework that decisively resolves this trade-off. By integrating an intelligent feature fusion mechanism within a highly optimized object detection architecture, the proposed model achieves state-of-the-art performance. Rigorous quantitative analysis demonstrates its superiority over previous models, exhibiting exceptional accuracy, precision, and recall, thereby providing a reliable system that minimizes the false alarm rates that have hindered past deployments. Furthermore, a detailed analysis of the model's performance under specific environmental stressors—including high wind, low thermal contrast, rain, snow, and fog—reveals its remarkable resilience to the most common causes of system failure. Finally, we have shown that the model is not merely a theoretical construct but has been engineered for practical application, with a clear pathway for optimization and deployment on resource-constrained edge computing hardware.

Looking forward, the development of this robust detection framework opens several promising avenues for future research and development.

1. **Integrate optical camera with OGI thermal camera:** To reduce the noise and false positive, we can provide optical RGB color view from the same angle as OGI camera and develop dual model that combine both footages to reduce FS.
2. **Quantitative OGI (QOGI):** The immediate next step is to extend the model's capabilities from binary detection to accurate leak rate quantification. Integrating the model's precise plume segmentation with physical models and telemetry data (e.g., wind speed, distance) can enable real-time, accurate rate estimation, addressing a critical industry need for emissions inventory reporting.
3. **Multi-Sensor Fusion:** The goal is a holistic leak detection and repair (LDAR) ecosystem. Future systems could integrate the real-time alerts from the OGI model with data from a network of complementary sensors, such as fixed-point laser detectors or High-resolution infrared emission monitoring of open tip flares to recognize gas from steam and vapor. This fusion of technologies would create a multi-layered, redundant system, providing unparalleled accuracy, reliability, and actionable intelligence for mitigating methane emissions across the oil and gas value chain.

In conclusion, the framework presented in this paper represents a significant advancement in the pursuit of safer, cleaner, and more efficient energy operations, providing a powerful new tool to help the industry meet the environmental challenges of the 21st century.

References

- Cusworth, D.H., Duren, R.M., Thorpe, A.K., Olson-Duvall, W., Heckler, J., Chapman, J.W., Eastwood, M.L., Helmlinger, M.C., Green, R.O., Asner, G.P. and Dennison, P.E., 2021. Intermittency of large methane emitters in the Permian Basin. *Environmental Science & Technology Letters*, 8(7), pp.567–573.

- Fox, T.A., Barchyn, T.E., Risk, D., Ravikumar, A.P. and Hugenholtz, C.H., 2019. A review of close-range and screening technologies for mitigating fugitive methane emissions in upstream oil and gas. *Environmental Research Letters*, **14**(5), p.053002.
- Intergovernmental Panel on Climate Change (IPCC). (2014). *Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. IPCC, Geneva, Switzerland.
- International Energy Agency (IEA). (2020). *Methane Tracker 2020*. IEA, Paris.
- Korjani, M., Conley, D., and Smith, M. (2024). Temporal Deep Learning Image Processing Model for Natural Gas Leak Detection Using OGI Camera. In *Offshore Technology Conference Asia*. OTC-34756-MS.
- Oil and Gas Methane Partnership (OGMP). (2020). *OGMP 2.0 Reporting Framework*. UNEP.
- Wang, J., Tchapmi, L.P., Ravikumar, A.P., McGuire, M., Bell, C.S., Zimmerle, D., Savarese, S., and Brandt, A.R. (2020). Machine vision for natural gas methane emissions detection using an infrared camera. *Applied Energy*, **257**, 113998.
- Wang, J., Ji, J., Ravikumar, A.P., Savarese, S., and Brandt, A.R. (2022). VideoGasNet: Deep learning for natural gas methane leak classification using an infrared camera. *Energy*, **238**, 121733.
- Zimmerle, D., T. Vaughn, C. Bell, K. Bennett, P. Deshmukh, and E. Thoma. (2020). Detection Limits of Optical Gas Imaging for Natural Gas Leak Detection in Realistic Controlled Conditions. *Environmental Science & Technology* **2020** **54** (18), 11506–11514